Truck APS Failure Prediction Using Machine Learning

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Introduction

Truck failures related to the Air Pressure System (APS) can lead to serious operational issues. This project uses machine learning models to predict whether a failure is APS-related using sensor data. The dataset is highly imbalanced (~1.67% failures), so the primary objective is to minimize false negatives (missed failures).

Data Set

- APS Failure Dataset (Scania Trucks)
- 60,000 samples, 171 features
- Binary classification:
  - pos = APS failure
  - neg = not APS

Challenges:
- Severe class imbalance
- Missing data across features

Methodology

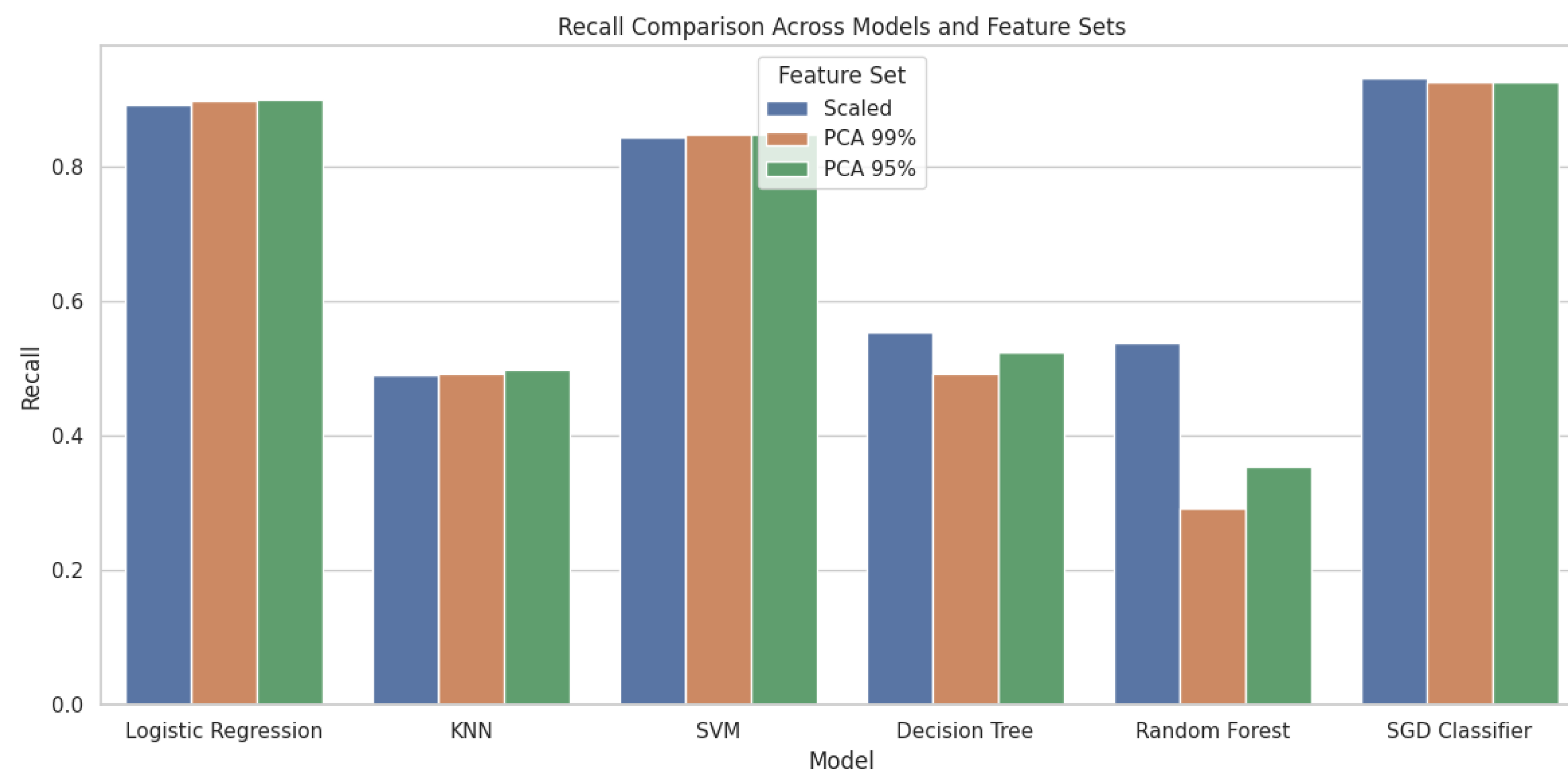
Preprocessing:
- Removed features with >70% missing values
- Median imputation
- Standard scaling

Models Tested:
- Logistic Regression
- KNN, SVM
- Decision Tree, Random Forest
- SGDClassifier

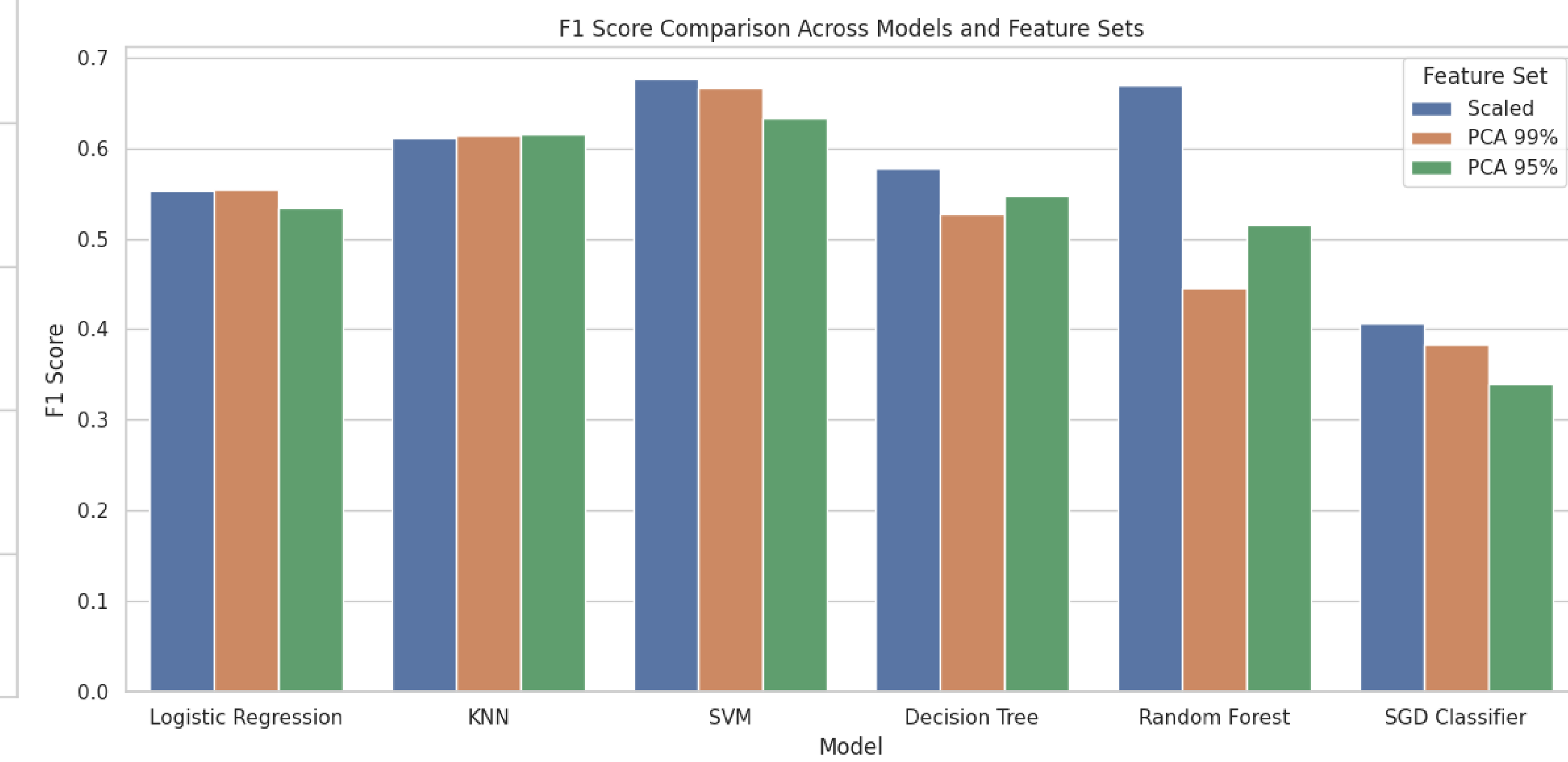
Model Selection:
- Cross-validation (5-fold)
- Metrics: Accuracy, Precision, Recall, F1
- Focus on Recall

Feature Engineering:
- PCA tested (99% & 95%)
- Minimal performance impact
- Not used in final model

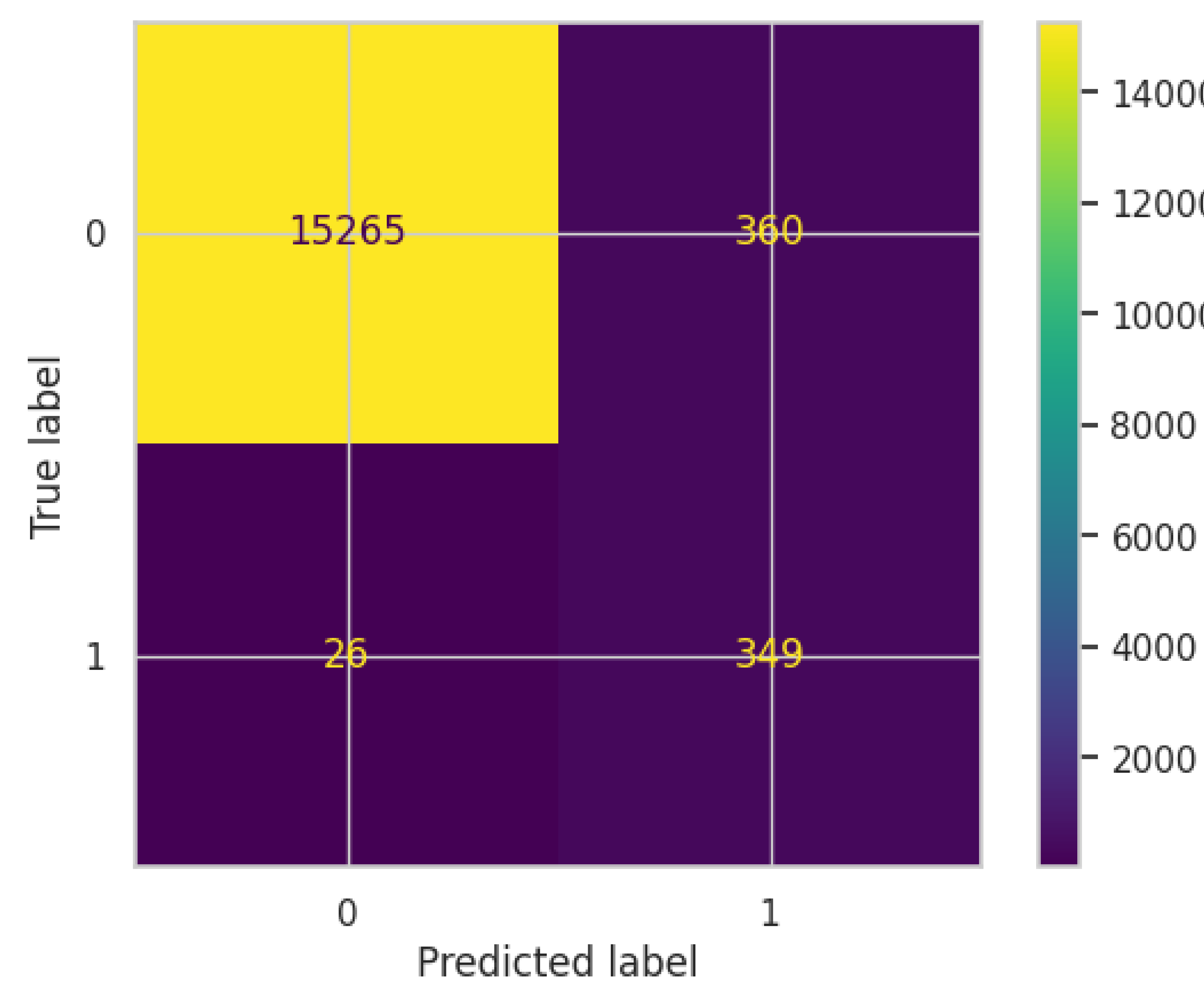
Recall model comparison



F1 model comparison



Logistic Regression confusion matrix



- True Positives: 349
- False Negatives: 26
- False Positives: 360

Detects most failures while maintaining a reasonable number of false positives.

Final model comparison

Model	Recall	Precision	F1
Logistic Regression	0.93	0.49	0.64
SGDClassifier	0.95	0.35	0.51

Results

Logistic Regression and SGDClassifier achieved the highest recall. While SGDClassifier detected slightly more failures, it produced significantly more false positives. Logistic Regression provided a better balance between recall and precision.

Conclusions

Logistic Regression was selected as the final model. It achieves high recall while maintaining better precision than SGDClassifier. This provides a practical balance between detecting failures and avoiding excessive false alarms..

Future work

Future Work:
- Apply SMOTE for class imbalance
- Explore feature selection
- Test gradient boosting models

References

APS Failure Dataset (Scania Trucks), 2016 Industrial Data Analysis Challenge.UCI Machine Learning Repository:https://archive.ics.uci.edu/dataset/414/ida2016challenge